Qi Huang

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EDUCATION

The University of Chicago — **Sept 2025 – Jun 2027**
M.Sc. in Computational and Applied Mathematics — *Chicago, US*

The University of Liverpool — **Sept 2023 – Jun 2025**
B.Sc. in Mathematics, First Class Honours · Ranked Top 1% in Major — *Liverpool, UK*

Xi'an Jiaotong-Liverpool University — **Sept 2021 – Jul 2023**
B.Sc. in Applied Mathematics — *Suzhou, China*

RESEARCH EXPERIENCE

Causal Inference for Observational Studies — **Nov 2025 – Present**
University of Chicago, Department of Statistics | *Mentor:* ***Prof. Xinran Li***

- Developed a covariate-adaptive studentized sensitivity analysis for Neyman's weak null in matched observational studies, anchoring the within-set assignment law at fitted propensity odds rather than the uniform law.
- Proved the proposed Horvitz–Thompson lower envelope is mean-controlled and pointwise sharp with asymptotically valid studentized tests; extended it in closed form to full matching and applied it to a right heart catheterization study.

Scientific World Models and LLM Agents — **Mar 2026 – Present**
Princeton University, Zhuang Lab | *Mentor:* ***Prof. Zhuang Liu***

- Building a scientific world model that predicts outcomes of candidate research actions (research state + action → predicted outcome) to guide autonomous experiment planning with agents and vision-language models.
- Designed mechanism-aware predictors and a test-time selective-adaptation router for out-of-distribution generalization across unseen action families; reduced planner regret by ~80% in held-out replay evaluations.

AI Breakthroughs and Intellectual Diversity — **Sept 2025 – Jul 2026**
University of Chicago, Knowledge Lab | *Mentor:* ***Prof. James A. Evans***

- Led mathematical modeling for a large-scale study of how AI breakthroughs concentrate scientific attention and erode intellectual diversity, drawing on 7.18M papers, 73K peer reviews, and 3.7M scientists across 36 years.
- Co-authored the resulting manuscript, "Stronger Artificial Intelligence Breakthroughs Hasten Their Own Exhaustion"; submitted to Science.

Audio-Visual Test-Time Adaptation — **Sept 2025 – Jun 2026**
University of Wisconsin–Madison | *Mentor:* ***Prof. Xiao Luo***

- Developed an audio-visual test-time adaptation framework under severe distribution shift using binary modality accept/reject gating rather than softmax-style weighting.
- Designed learned attention maps with entropy-adaptive token dropout, curriculum annealing, and class-specific thresholds to reduce overfitting to noisy pseudo-labels.

SELECTED PUBLICATIONS AND WORKING PAPERS

- **Qi Huang***, Pengyun Wang*, Xinran Li. *Covariate-Adaptive Studentized Sensitivity Analysis for Neyman's Weak Null in Matched Designs*. Working paper (*equal contribution)
- Pengyun Wang, **Qi Huang**, Jia Huang, Yutong Zhou. *Simultaneous Inference for Proximal Causal Learning with Weak Proxies*. Under review, International Conference on Learning Representations (ICLR) 2027
- Pengyun Wang, Yutong Zhou, **Qi Huang**. *Robust Propagation-Integrated Inference for Observational Studies with Network Interference*. Under review, International Conference on Learning Representations (ICLR) 2027
- Yuxuan Hu, Yushan Pan, Minghui Zhu, Xiaojiang He, Nan Xiang, **Qi Huang**, Ruizhi Ning, Yanlong Cao, Zhijie Xu. *QEGF: A Quantum-Inspired Geometric and Energy-Based Fusion Framework for Multimodal Sentiment Analysis*. Accepted, IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2026
- Pengyun Wang, Junyu Luo, Yanxin Shen, Ming Zhang, Shaoen Qin, Hanwen Xing, Siyu Heng, **Qi Huang**, Xiao Luo. *A Comprehensive Graph Pooling Benchmark: Effectiveness, Robustness and Generalizability*. Major revision, ACM Transactions on Knowledge Discovery from Data (TKDD)
- Jia Huang, Hongyou Zhou, **Qi Huang**, Hongbin Liu, Peng Zhang, Zhengqi Sun, Lumin Xu. *PVI-DPO: Perceptual Video Inpainting via Informative-Safe Preference Mining and Region-Aware Direct Preference Optimization*. Submitted to AAAI Conference on Artificial Intelligence (AAAI) 2027
- Beichen Lu, Honglin Bao, **Qi Huang**, Qingguo Meng, James A. Evans. *Stronger Artificial Intelligence Breakthroughs Hasten Their Own Exhaustion*. Submitted to Science

INTERNSHIPS

iFLYTEK **Jun 2025 – Sept 2025**
Machine Learning Research Intern *Hefei, China*

- Trained and evaluated Transformer baselines in PyTorch on in-house text corpora; improved BLEU by +2.3 (27.6 → 29.9) through regularization and decoding tuning.
- Trained Conformer baselines on internal speech data; reduced WER from 10.3% to 8.9% via LR scheduling, SpecAugment, and label smoothing.

Alibaba Group **Jul 2024 – Aug 2024**
Data Analyst Intern, Finance Department *Hangzhou, China*

- Cleaned and analyzed consumer data using AARRR metrics; supported retention analysis and precision marketing with RFM-based customer segmentation.
- Built activity and retention visualizations in Matplotlib and Pandas to support management decision-making.

ACADEMIC SERVICE

Reviewer: Forecast@ICML 2026 Workshop (Forecasting as a New Frontier of Intelligence)

SKILLS

Programming: Python, SQL, R, MATLAB, Java, VBA, Maple
Tools: PyTorch, Pandas, Matplotlib, Tableau, Power BI, Jupyter Notebook, LaTeX, Excel